

Phase 1 Static FJSP Mathematical Model

FJSP-AGV-RL Research Platform

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1 Scope

This document describes the Phase 1 static flexible job shop scheduling problem (FJSP) model implemented by the repository. Transportation time, AGV routing, dynamic job arrivals, reinforcement learning, and graph neural network components are outside the Phase 1 scope.

2 Sets and Indices

Symbol	Meaning
J	set of jobs, indexed by j
O_j	ordered set of operations for job j , indexed by o
M	set of machines, indexed by m
A_{jo}	eligible machines for operation (j, o)

Operation (j, o) must precede $(j, o + 1)$ for every job j and operation position o where $o + 1 \in O_j$.

3 Parameters

Symbol	Meaning
p_{jom}	processing time of operation (j, o) on machine m
H	conservative scheduling horizon

The implementation uses a conservative horizon

$$H = \sum_{j \in J} \sum_{o \in O_j} \max_{m \in A_{jo}} p_{jom}.$$

This upper bound is not necessarily tight, but it is safe for the small CP-SAT baseline model.

4 Decision Variables

Here a and b denote operation identifiers such as (j, o) . The explicit y variables are useful for a classical mixed-integer programming statement. The current OR-Tools implementation uses optional interval variables and `NoOverlap` instead of manually creating all y variables.

Symbol	Meaning
$x_{jom} \in \{0, 1\}$	1 if operation (j, o) is assigned to machine m
$S_{jo} \geq 0$	start time of operation (j, o)
$C_{jo} \geq 0$	completion time of operation (j, o)
$C_{\max} \geq 0$	makespan
$y_{abm} \in \{0, 1\}$	order variable for two operations a, b on machine m

5 Objective

The objective is to minimize the makespan:

$$\min C_{\max}.$$

6 Constraints

6.1 Machine Assignment

Each operation must be assigned to exactly one eligible machine:

$$\sum_{m \in A_{jo}} x_{jom} = 1, \quad \forall j \in J, o \in O_j.$$

Machines outside A_{jo} are not valid choices and are never created as candidate assignments in the implementation.

6.2 Processing Time Link

If operation (j, o) is assigned to machine m , then its completion time must match its start time plus the selected processing time:

$$C_{jo} \geq S_{jo} + p_{jom} - H(1 - x_{jom}), \quad \forall j, o, m \in A_{jo}.$$

6.3 Job Precedence

Operations within the same job must follow the given route order:

$$C_{jo} \leq S_{j,o+1}, \quad \forall j \in J, o, o+1 \in O_j.$$

6.4 Machine Capacity

A machine can process at most one operation at a time. For any two distinct operations $a = (j, o)$ and $b = (k, q)$ that are both eligible for machine m , the disjunctive ordering is:

$$S_a + p_{am} \leq S_b + H(1 - y_{abm}) + H(2 - x_{am} - x_{bm}),$$

$$S_b + p_{bm} \leq S_a + H y_{abm} + H(2 - x_{am} - x_{bm}).$$

These constraints are active only when both operations are assigned to machine m . In the CP-SAT implementation, this logic is represented by optional intervals and a per-machine NoOverlap constraint.

6.5 Makespan Definition

The makespan must be no smaller than every final operation completion time:

$$C_{\max} \geq C_{j, \text{last}(j)}, \quad \forall j \in J.$$

7 Feasibility Checker Mapping

The repository’s feasibility checker validates the following model conditions:

- every operation appears exactly once;
- all job and operation ids are inside the instance range;
- every selected machine is eligible for the operation;
- recorded processing time matches the instance data;
- job precedence constraints are satisfied;
- no two operations overlap on the same machine;
- stored makespan equals the maximum recorded completion time.

8 Dispatching Rules

The strict dispatching rules use a common candidate tuple

$$(j, o, m, r_{jo}, s_{jom}, f_{jom}, p_{jom}),$$

where r_{jo} is the job ready time, s_{jom} is the earliest feasible start time on machine m , and $f_{jom} = s_{jom} + p_{jom}$ is the tentative finish time.

Rule	Operation selection	Machine selection
FIFO	smallest (r_{jo}, j, o)	earliest finish
SPT	smallest p_{jom} pair	selected with pair
EFT	smallest f_{jom} pair	selected with pair
MWKR	largest remaining route work	earliest finish
Random	random ready pair	selected with pair

For MWKR, remaining route work is estimated as

$$W_{jo} = \sum_{q=o}^{last(j)} \min_{m \in A_{jq}} p_{jqm}.$$

9 Experiment Result Schema

Phase 1 experiment scripts write normalized CSV files under `experiments/results/`. Each row has the following fields:

The CSV layer is intentionally simple. It records what happened in each experiment without changing solver behavior or feasibility definitions.

Field	Meaning
<code>experiment_id</code>	stable experiment identifier
<code>instance_name</code>	benchmark or toy instance name
<code>algorithm</code>	solver or dispatching rule name
<code>makespan</code>	objective value, empty if no schedule was returned
<code>feasible</code>	PASS or FAIL after feasibility checking
<code>runtime_s</code>	measured wall-clock runtime in seconds
<code>note</code>	solver status, error, or validation note